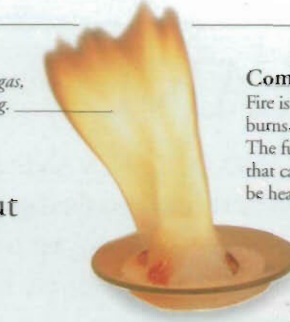
FIRE

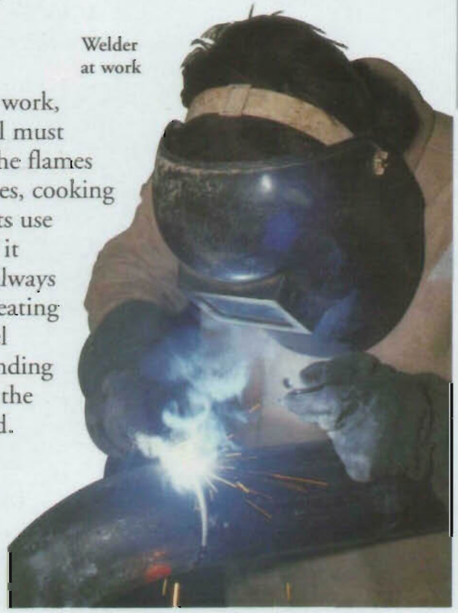
BURNING OUT OF CONTROL in forests or cities, a fire leaves a trail of destruction. Yet life without the benefits of fire is unimaginable. We use fire in power stations, car engines, and kitchens, to provide electricity and transport, or to cook food. Early humans realized the value of fire about half a million years ago – perhaps when lightning set a tree on fire. Learning to control and use the flames helped them hunt, clear land for farming, survive in colder climates, and eat foods that were inedible when raw. No wonder some religions still worship fire as a hungry god.



Combustion or burning
Fire is the heat and light produced when fuel burns. This process is known as combustion. The fuel can be any flammable material (one that can catch fire). The material must first be heated to a temperature called the ignition temperature; above this, it will burst into flame. As a fire gets hotter, more fuel catches alight, and the flames spread. Gases and vapours burn quickly; liquids and solids take longer to burn.

Using fire

To make fire do useful work, the supply of air or fuel must be controlled to keep the flames burning evenly. Furnaces, cooking stoves, and power plants use fire for the heat energy it produces. Heat is not always the main purpose of creating fire. In a car engine fuel burns explosively. Expanding gases drive the vehicle; the heat produced is wasted.



Welder at work

Welding
Many industrial processes rely on combustion. In the welder's torch, oxygen and acetylene gas mix and produce a flame hot enough to melt steel.

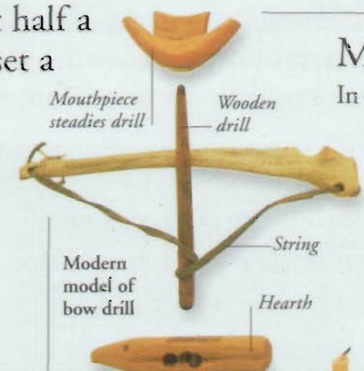
Cooking
Many foods must be cooked before they can be eaten. When food is heated, chemical changes take place that improve its taste and make it easier to digest. Early people ate raw food until they discovered cooking, probably by accident.

Cooking with fire

Fire-engine with hydraulic platform, used to reach awkward spaces.

Making a fire

In the past, there were two main methods of starting a fire: raising the temperature until flames appeared, or striking sparks to set light to tinder. Cigarette lighters still start fires by using the spark of flint on steel.



Modern model of bow drill

Bow drills
Rapidly turning the string of a bow drill causes friction at the tip, which starts flames.

A tinder box contains flint, which makes sparks when struck against metal (the steel).

A fire piston works like a bicycle pump: compressing air in the tube raises the temperature until the tinder (flammable material) inside catches alight.

Matches
Invented in 1827, these wooden splinters were tipped with chemicals. The chemicals were ignited by heat, generated by rubbing the tip against sandpaper. Safety matches burn only when rubbed against a specially coated strip on the matchbox.

Fighting fire

Fires feed on fuel, air, and heat; removing any one of these puts out the flames. Firefighters spray a blaze with water to remove heat and to create a blanket of steam that chokes off the air supply.

Myths about fire

The power and danger of fire made ancient peoples wonder about its origin. Myths that explain how people learned to tame flames occur in many separate cultures. Most fire myths involve a hero who brings fire to the world.

Prometheus
In Greek mythology, the chief god, Zeus, hid the secret of fire from mortals (humans) to punish them for a trick that a lesser god, Prometheus, had played on him. But Prometheus snatched a glowing ember from the Sun, and brought fire to the Earth.

Prometheus

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